LM03 Derivative Benefits, Risks, and Issuer and Investor Uses

Version 1.0

1. Introduction

This learning module covers:

- The benefits and risks of using derivatives
- A comparison of how issuers use derivatives versus how investors use derivatives

2. Derivative Benefits

Some of the benefits of derivatives are listed below:

Risk Allocation, Transfer, and Management

Derivative instruments enable users to allocate, transfer, and/or manage risk without trading an underlying.

In many scenarios, issuers and investors face a timing difference between making an economic decision and being able to transact in a cash market. The curriculum provides the following examples:

Issuers:

- A manufacturer may need to order commodity inputs for its production process in advance of receiving finished-goods orders.
- A retailer may await a shipment of goods priced in a foreign currency before selling domestic currency to make payment.
- An issuer may wish to lock in its future debt costs in advance of the maturity of an outstanding debt issuance.

Investors:

- An investor may seek to capitalize on a market view but lack the necessary cash on hand to transact in the spot market.
- In anticipation of a future stock dividend, debt coupon, or principal repayment, an investor may decide today how it will reinvest the proceeds in the future.

In such scenarios, derivatives can be used to lock in a pre-agreed price for a future date and bridge the timing gap. For example, if an investor has a bullish view on the market but lacks the cash to transact in the spot market, he can purchase a forward contract or a call option on an index to benefit from his view.

Derivatives allow users to create exposure profiles which are unavailable in cash markets. For example, a short call position can be added to an existing long cash position to create a 'covered call' position. A covered call generates a higher return as compared to the original long cash position if the underlying price is stable or slightly higher.

Instructor's Note: This concept is covered in a later learning module.

Information Discovery

There are two primary advantages of futures markets:

- Price discovery: Market participants often use futures prices to forecast the direction of cash markets. For example, analysts frequently use interest rate futures markets to gauge investor expectations of a central bank's benchmark interest rate increase or decrease at a future meeting.
- Implied volatility: Implied volatility measures the expected risk of the underlying. With the models such as BSM to price an option, it is possible to determine the implied volatility, and hence the risk.

  Instructor's Note: This concept is covered in a later learning module.

Operational Advantages

Some of the major operational advantages associated with derivatives are given below:

- Lower transaction costs than the underlying.
- Greater liquidity than the underlying spot markets.
- Margin requirements and option premiums are low relative to the cost of the underlying
- Easy to take a short position.

Market Efficiency

Derivatives markets offer a less costly way to exploit mispricings in the market due to operational advantages: low transaction costs, easier to take a short position, etc. Therefore, any mispricing is corrected more quickly in the derivatives market than the spot market.

Thus, the existence of derivative markets causes financial markets in general to function more efficiently.

Exhibit 1 from the curriculum summarizes the benefits of derivative instruments.

Purpose	Description
Risk Allocation, Transfer, and Management	Allocate, trade, and/or manage underlying exposure without trading the underlying Create exposures unavailable in cash markets
Information Discovery	Deliver expected price in the future as well as expected risk of underlying
Operational Advantages	Reduced cash outlay, lower transaction costs versus the underlying, increased liquidity and ability to "short"
Market Efficiency	Less costly to exploit arbitrage opportunities or mispricing

3. Derivative Risks

The main risks associated with derivatives are listed below:

Greater Potential for Speculative Use

Derivatives have a high degree of implicit leverage as compared to a similar cash position. This magnifies profits/losses and increases the likelihood of financial distress.

Lack of Transparency

Derivatives increase a portfolio's complexity. They may create an exposure profile that is not well understood by stakeholders.

This risk increases when a combination of derivatives and/or embedded derivatives is used. For example, structured notes are a broad category of securities that combine features of debt instruments with embedded derivatives designed to achieve a particular issuer or investor objective. They often involve greater cost, lower liquidity, and less transparency as compared to an equivalent stand-alone instrument.

Basis Risk

Derivatives are often used to hedge the risk of a commercial or financial exposure. The assumption here is that the derivative instrument will be highly effective in offsetting the price risk of the underlying. However, in some cases, the expected value of a derivative differs unexpectedly from the underlying. This is referred to as 'basis risk'.

Basis risk can occur when a derivative instrument refers to a price or index that is similar to, but does not exactly match, an underlying exposure.

Liquidity Risk

Liquidity risk refers to the divergence in cash flow timing of a derivative versus that of an underlying transaction. For example, future contracts require daily settlement of gains and losses and can give rise to liquidity risk. An investor using futures to hedge an underlying transaction may be unable to meet a margin call due to lack of funds, and may be forced to close out his position.

Counterparty Credit Risk

Derivatives can result in significant counterparty credit exposure. Unlike bond markets where credit exposures are easy to predict based on the notional principal and accrued interest; predicting the credit exposures of derivatives is more challenging. Credit exposures vary based on the type of derivatives. For example, a call option seller has no counterparty credit risk exposure after receiving the premium, but a call option buyer is exposed to counterparty credit risk, which varies depending on the price of the underlying.

Destabilization and Systemic Risk

Derivatives are often blamed to have destabilizing consequences on the financial markets. This is primarily due to the high amount of leverage taken by speculators. If the position turns against them, then they default. This triggers a ripple effect causing their creditors to default, their creditors' creditors to default, and so on. A default by speculators impacts the whole system. For example, the credit crisis of 2008.

Many market reforms such as the central clearing mandate for swaps between financial intermediaries and a central counterparty (CCP), have been put in place to reduce systemic risk.

Exhibit 3 from the curriculum summarizes the key risks associated with derivative instruments and markets.

Risk	**Description**
Greater Potential for Speculative Use	High degree of implicit leverage for some derivative strategies may increase the likelihood of financial distress.
Lack of Transparency	Derivatives add portfolio complexity and may create an exposure profile that is not well understood by stakeholders.
Basis Risk	Potential divergence between the expected value of a derivative instrument versus an underlying or hedged transaction
Liquidity Risk	Potential divergence between the cash flow timing of a derivative instrument versus an underlying or hedged transaction
Counterparty Credit Risk	Derivative instruments often give rise to counterparty credit exposure, resulting from differences in the current price versus the expected future settlement price.
Destabilization and Systemic Risk	Excessive risk taking and use of leverage in derivative markets may contribute to market stress, as in the 2008 financial crisis.

4. Issuer Use of Derivatives

Issuers typically use derivative instruments to offset or hedge market-based underlying exposures that impact their assets, liabilities, and earnings. For example, companies often use FX forward contracts to hedge FX exposures related to import/export transactions.

Issuers also seek hedge accounting, which allows an issuer to offset a hedging instrument (usually a derivative) against a hedged transaction or balance sheet item. This reduces income statement and cash flow volatility.

Hedge types include:

- **Cash flow hedge**: A derivative designated as absorbing the variable cash flow of a floating-rate asset or liability. For example, FX forwards are a cash flow hedge that offset the variability of exchange rates.
- **Fair value hedge**: A derivative used to offset the fluctuation in fair value of an asset or liability. For example, a commodities producer anticipating lower future prices may use forward contracts to sell its inventory and lock-in a price.
- **Net investment hedge**: A net investment hedge occurs when either a foreign currency bond or a derivative such as an FX swap or forward is used to offset the exchange rate risk of the equity of a foreign operation.

To qualify for hedge accounting, the dates, notional amounts, and other contract features of a derivative must closely match those of the underlying transaction. Therefore, issuers are more likely to use OTC derivatives customized to meet their specific needs rather than exchange traded derivatives.

5. Investor Use of Derivatives

Investors use derivatives to:

- replicate a cash market strategy (e.g. purchasing a futures contract on a stock to replicate its cash market performance)
- hedge a fund's value against adverse movements in underlyings (e.g. purchasing put options to protect an equity portfolio from significant losses during an uncertain event)
- modify or add exposures using derivatives which may be unavailable in cash markets (e.g. covered call)

The prospectus of an investment fund usually specifies which derivative instruments may be used within the fund and for what purpose.

Investors are typically less concerned with hedge accounting than issuers. As a result, they tend to trade in standardized and highly liquid exchange-traded derivatives more frequently than issuers.

Summary

LO. Describe benefits and risks of derivative instruments.

Benefits

Purpose	Description
Risk Allocation, Transfer, and Management	Allocate, trade, and/or manage underlying exposure without trading the underlying Create exposures unavailable in cash markets
Information Discovery	Deliver expected price in the future as well as expected risk of underlying
Operational Advantages	Reduced cash outlay, lower transaction costs versus the underlying, increased liquidity and ability to "short"
Market Efficiency	Less costly to exploit arbitrage opportunities or mispricing

Risks

Risk	Description
Greater Potential for Speculative Use	High degree of implicit leverage for some derivative strategies may increase the likelihood of financial distress.
Lack of Transparency	Derivatives add portfolio complexity and may create an exposure profile that is not well understood by stakeholders.
Basis Risk	Potential divergence between the expected value of a derivative instrument versus an underlying or hedged transaction
Liquidity Risk	Potential divergence between the cash flow timing of a derivative instrument versus an underlying or hedged transaction
Counterparty Credit Risk	Derivative instruments often give rise to counterparty credit exposure, resulting from differences in the current price versus the expected future settlement price.
Destabilization and Systemic Risk	Excessive risk taking and use of leverage in derivative markets may contribute to market stress, as in the 2008 financial crisis.

LO. Compare the use of derivatives among issuers and investors.

Issuers typically use derivative instruments to offset or hedge market-based underlying exposures that impact their assets, liabilities, and earnings. Issuers also seek hedge accounting, which allows an issuer to offset a hedging instrument (usually a derivative) against a hedged transaction or balance sheet item. This reduces income statement and cash flow volatility.

Investors use derivatives to replicate a cash market strategy, hedge a fund's value against adverse movements in the underlying, modify or add exposures using derivatives which may be unavailable in cash markets.